Arista Access Node WFD-ACC-48 Routing Protocol Flap Resolution SOP

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5362

Description

The routing protocol flap failure on Arista Access Node WFD-ACC-48 located at Whitefield POP is primarily due to the dynamic nature of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. In this context, the Access Node is configured to operate in a high-availability environment, where multiple links are provisioned to ensure continuous service availability. However, when a link failure occurs, it triggers a cascade of routing protocol events, leading to a flap in the routing information base (RIB) of the Access Node. This flap results in a major alarm (ALARM_5362) being raised, indicating a critical failure of the routing protocol convergence mechanism. To mitigate this failure, it is essential to identify and rectify the root cause of the flap, which could be attributed to various factors including misconfiguration, hardware failure, or external network instability.

Pre-requisite Checks

- check: Verify redundant link status on WFD-ACC-48 Access Node
- check: Confirm power and cooling status at Whitefield POP
- check: Ensure remote hands are dispatched to Whitefield POP for immediate assistance

Potential Root Causes

Cause	Probability
Misconfiguration of BGP routing policy	High
Hardware failure of WFD-ACC-48 Access Node	Medium
External network instability affecting Whitefield POP	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 10 Displayed 1(1/0) Output queue 0, 0 dropped Non-stop routing is enabled, Description: This command displays an overview of the BGP routing table and its version, helping to identify any recent updates or changes that may be contributing to the flap.

Command: show ospf database, Expected Output: OSPF database for process 1.0.0.0 OSPF database (Sequence#: 0) OSPF Database (Type: 1) OSPF Database (Type: 2) OSPF Database (Type: 3), Description: This command provides a snapshot of the OSPF database, which can help detect any adjacency loss or other issues that may be contributing to the flap.
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```
Command: show isis databases, Expected Output: ISIS database for process Level-2 ISIS Level-2 database (Sequence#: 0) ISIS Level-2 database (Type: 1) ISIS Level-2 database (Type: 2), Description: This command displays the ISIS database, helping to identify any LSP flooding or other issues that may be contributing to the flap.
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Resolution Steps

Step 1: Step: Verify BGP routing policy configuration, Command: show running-config | section bgp, Description: Verify that the BGP routing policy configuration is correct and matches the expected behavior. Ensure that the AS_PATH and COMMUNITY attributes are properly set and do not contain any errors.

Step 2: Step: Clear BGP routing information, Command: clear ip bgp *, Description: Clear the BGP routing information to reset the routing table and resolve any inconsistencies.

Step 3: Step: Verify OSPF adjacency status, Command: show ip ospf neighbor, Description: Verify the OSPF adjacency status to ensure that all neighbors are connected and communicating properly.

Step 4: Step: Clear OSPF routing information, Command: clear ip ospf database, Description: Clear the OSPF routing information to reset the routing table and resolve any inconsistencies.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 10 Displayed 1(1/0) Output queue 0, 0 dropped Non-stop routing is enabled, Description: Verify that the BGP routing table has converged and the flap has been resolved.

Command: show ospf database, Expected Output: OSPF database for process 1.0.0.0 OSPF database (Sequence#: 0) OSPF Database (Type: 1) OSPF Database (Type: 2) OSPF Database (Type: 3), Description: Verify that the OSPF database has converged and the adjacency loss has been resolved.

Command: show isis databases, Expected Output: ISIS database for process Level-2 ISIS Level-2 database (Sequence#: 0) ISIS Level-2 database (Type: 1) ISIS Level-2 database (Type: 2), Description: Verify that the ISIS database has converged and the LSP flooding has been resolved.
```

Roll-back Procedures

- Step: Restore previous BGP routing policy configuration, Command: copy running-config startup-config, Description: Restore the previous BGP routing policy configuration to resolve any changes made during the resolution process.

- Step: Clear BGP routing information, Command: clear ip bgp *, Description: Clear the BGP routing information to reset the routing table and resolve any inconsistencies.

- Step: Verify OSPF adjacency status, Command: show ip ospf neighbor, Description: Verify the OSPF adjacency status to ensure that all neighbors are connected and communicating properly.

- Step: Clear OSPF routing information, Command: clear ip ospf database, Description: Clear the OSPF routing information to reset the routing table and resolve any inconsistencies.